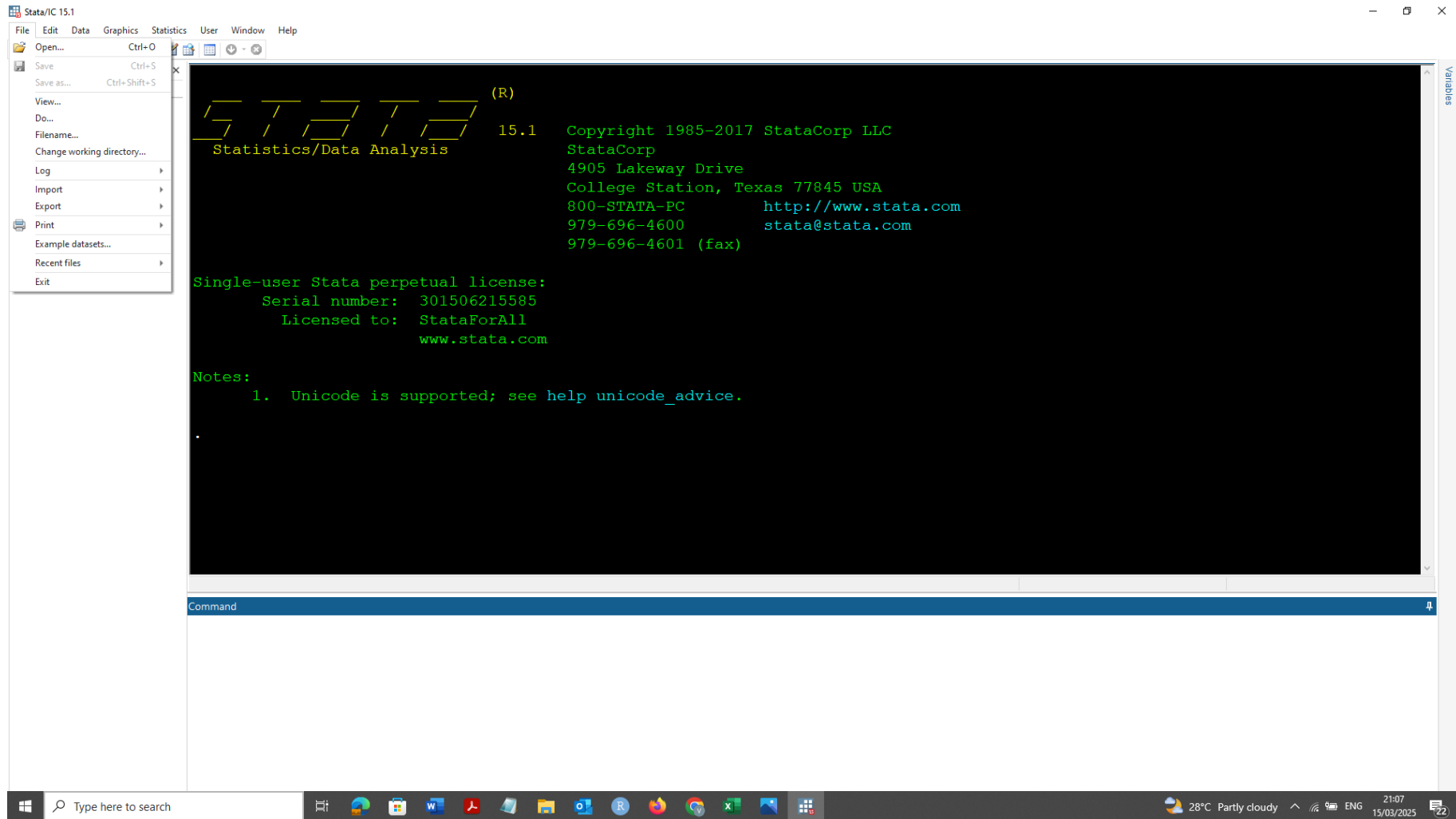
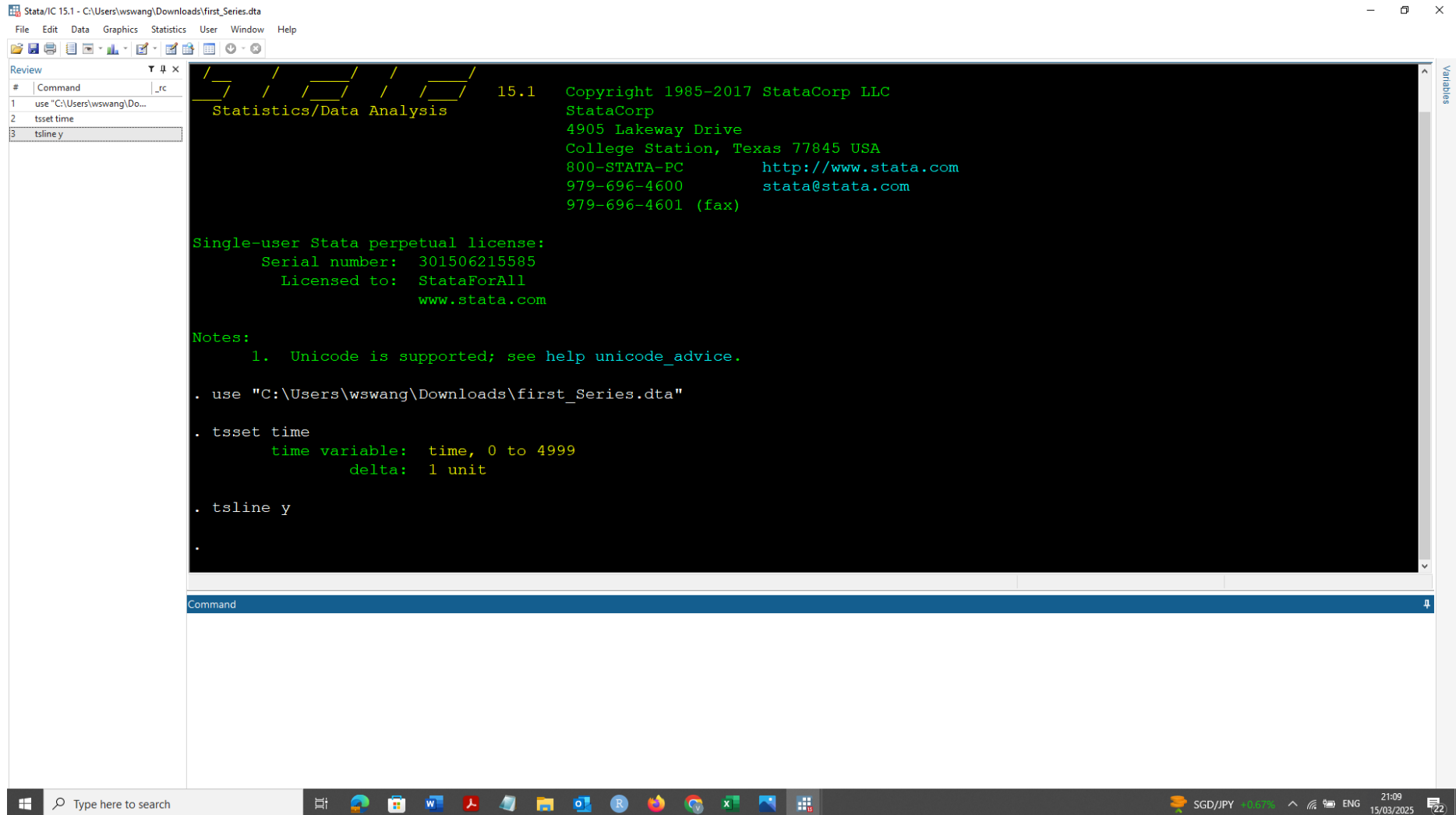


1

(a) We can open the data set (first series) through the open option.



Let Stata know that it is a time series dataset using 'tsset time'. Then use 'tsline' to create the time series line.



The screenshot shows the Stata 15.1 interface. The main window displays the Stata startup screen with the Stata logo and version information. The command window on the left shows the following commands:

```
# Command
1 use "C:\Users\wswang\Downloads\first_Series.dta"
2 tsset time
3 tsline y
```

The command window output shows the Stata startup screen with the following text:

```

      _ _ _ _ _
      / / / / /
      \ \ \ \ \
      Statistics/Data Analysis

15.1 Copyright 1985-2017 StataCorp LLC
      StataCorp
      4905 Lakeway Drive
      College Station, Texas 77845 USA
      800-STATA-PC http://www.stata.com
      979-696-4600 stata@stata.com
      979-696-4601 (fax)

Single-user Stata perpetual license:
      Serial number: 301506215585
      Licensed to: StataForAll
                  www.stata.com

Notes:
      1. Unicode is supported; see help unicode_advice.

. use "C:\Users\wswang\Downloads\first_Series.dta"

. tsset time
      time variable: time, 0 to 4999
      delta: 1 unit

. tsline y

.
```

The command window output also shows the Stata startup screen with the following text:

```

      _ _ _ _ _
      / / / / /
      \ \ \ \ \
      Statistics/Data Analysis

15.1 Copyright 1985-2017 StataCorp LLC
      StataCorp
      4905 Lakeway Drive
      College Station, Texas 77845 USA
      800-STATA-PC http://www.stata.com
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      979-696-4601 (fax)

Single-user Stata perpetual license:
      Serial number: 301506215585
      Licensed to: StataForAll
                  www.stata.com

Notes:
      1. Unicode is supported; see help unicode_advice.

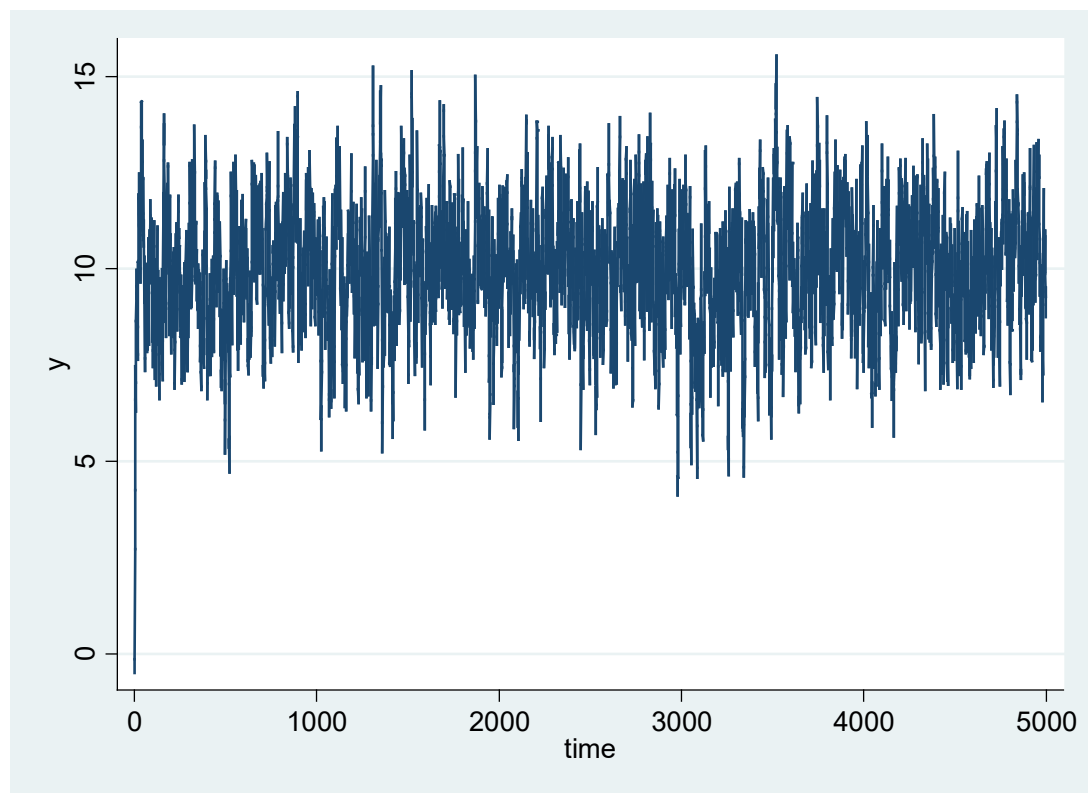
. use "C:\Users\wswang\Downloads\first_Series.dta"

. tsset time
      time variable: time, 0 to 4999
      delta: 1 unit

. tsline y

.
```

We can then quickly check the diagram to see whether the series is stationary. It appears that the series is more likely stationary because it has a constant mean (around 10) over time, and the variance is relatively stable over time, i.e., it fluctuates between 5 and 15 most of the time. However, to confirm its stationarity, we should use the unit root tests. Please note that for the stationary series, the covariance between $y(t)$ and $y(t-h)$ does not depend on t and cannot be checked from the diagram.



PP test

```
. pperron y
```

Phillips-Perron test for unit root

Number of obs = 4999
Newey-West lags = 9

	Test Statistic	Interpolated Dickey-Fuller		
		1% Critical Value	5% Critical Value	10% Critical Value
Z(rho)	-1011.765	-20.700	-14.100	-11.300
Z(t)	-24.042	-3.430	-2.860	-2.570

MacKinnon approximate p-value for Z(t) = 0.0000

ADF test

```
. dfuller y, lags(9)
```

Augmented Dickey-Fuller test for unit root

Number of obs = 4990

	Test Statistic	Interpolated Dickey-Fuller		
		1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-15.817	-3.430	-2.860	-2.570

MacKinnon approximate p-value for Z(t) = 0.0000

Reject the null hypothesis that
the series is nonstationary

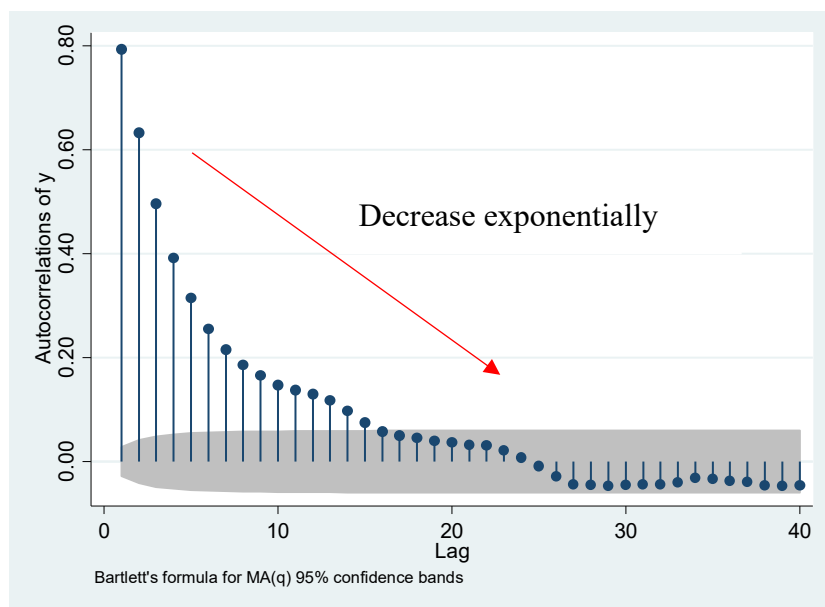
Note that, for the ADF test, I used lags (9) based on the suggestion from the PP test.

(b) Check autocorrelation function (ac). It appears that the bars drop to zero (i.e., the bars are in the grey area) quickly, unlike the McDonald's example discussed in class. If a series is nonstationary, the bars do not die down fast.

```
. tsset time
      time variable:  time, 0 to 4999
                delta: 1 unit

. tsline y

. ac y
```



(C)

```
. reg y l1.y
```

Source	SS	df	MS	Number of obs = 4999		
Model	8372.15451	1	8372.15451	F(1, 4997) = 8706.31		
Residual	4805.21145	4997	.961619262	Prob > F = 0.0000		
Total	13177.366	4998	2.6365278	R-squared = 0.6353		
				Adj R-squared = 0.6353		
				Root MSE = .98062		

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
y L1.	.7938323	.0085077	93.31	0.000	.7771534	.8105111
_cons	2.058418	.085992	23.94	0.000	1.889836	2.227

The estimated regression model is

$$\hat{y}_t = 2.05 + 0.793y_{t-1}$$

As we can see here, the estimated intercept is 2.05, which is very close to the true intercept value of 2, and the estimate slope 0.7938 is also very close to the true slope value of 0.8.

(d) Once we have done the estimation in (c), we can calculate the residuals and check if the residuals are white noise.

```
. predict uhat, r  
(1 missing value generated)
```

```
. wntestq uhat
```

Portmanteau test for white noise

Portmanteau (Q) statistic =	39.6316
Prob > chi2(40) =	0.4867

The null is that the residuals are white noise, and the alternative is the residuals are not white noise. Since the p value is 0.48 which is much larger than the 5% significance level, we cannot reject the null.

Remark:

As discussed during the tut, if the residuals are white noise, it implies Assumptions TS 4 and 5 hold.